Gregorio Zaltzman D'Ambrosio

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PROFILE

Trilingual Aerospace Engineering senior and dual EU (Italy)/Argentine citizen with experience spanning aircraft design, CFD, aerodynamics, structures, and propulsion. Combines analytical engineering rigor with hands-on leadership in multidisciplinary aerospace design-build-test projects.

EDUCATION

Technical University of Munich (TUM) — Munich, Germany | Oct 2026 – Aug 2028
M.Sc. Aerospace Engineering | Astronautical focus

University of California, San Diego — San Diego, CA, USA | Sep 2022 – Jun 2026
B.S. Aerospace Engineering | GPA: 3.3 Coursework: Aerodynamics, Aerospace Structures, Propulsion, Orbital Mechanics, Linear Control, Fluid Dynamics, Thermodynamics, Solid Mechanics, Numerical Methods, Engineering Experimental Techniques, Signals & Systems, Computational Fluid Dynamics, Spacecraft Guidance & Navigation

British School of Paris — Paris, France | Sep 2017 – Jun 2022
A-Levels: A* Mathematics, A Physics, A* EPQ | SAT: 780M / 660E | Duke of Edinburgh Gold Award

SKILLS

Analysis & Simulation	MATLAB, Python, JAX, IPOPT, ANSYS Fluent, XFLR5, XFOIL, AVL, OpenVSP, STK, CFD
Design & CAD	SolidWorks, Fusion 360, technical drawings, assemblies
Hardware & Lab	Arduino IDE, sensor instrumentation, data logging, rapid prototyping
Languages	English, Spanish, French (all fluent)

ENGINEERING PROJECTS

Blended Wing Body Commercial Airliner Conceptual Design — Chief Engineer — Winter 2026

- Led a 6-person team designing a 250-passenger Blended Wing Body for Mach 0.85 and 7,400 nmi.
- Maximized cruise L/D to 25.3 using Python Particle Swarm Optimization under yield and stability constraints.
- Defined the concept of operations, designed the 3 class cabin layout and a comprehensive cost analysis including Direct Operating Cost, Cost-Per-Available-Seat-Mile and a Break Even analysis

Payload Delivery RC Aircraft Build — MDO, CAD & AIT — Spring 2026

- Designed and fabricated an RC aircraft carrying a 7.22 L payload volume and ∼5 N payload weight.
- Developed a multidisciplinary design optimization workflow in Python using JAX and IPOPT to maximize mission profit.
- Manufactured carbon-fiber reinforced wings, laser-cut fuselage, and custom twin-bay payload deployment system.
- Conducted ground roll, propeller thrust, structural load, and payload deployment testing to validate takeoff performance, aerodynamic models, structural safety margins, and release reliability under operational conditions.

Vortex-Cooled Rocket Engine CFD Simulation — ANSYS Fluent — Spring 2026

- Performed compressible RANS and Conjugate Heat Transfer simulations of a vortex-cooled rocket engine.
- Quantified vortex insulation effectiveness across multiple oxidizer flow conditions and validated grid convergence.

Clark Y-14 Wind Tunnel Aerodynamic Validation — Winter 2026

- Measured lift, drag, and pitching moment characteristics using pressure taps, wake surveys, and force balances.
- Compared experimental data against XFLR5 predictions, identifying stall onset discrepancies.

Wing Spar Structural Analysis — MATLAB Tool — Winter 2025

- Developed a MATLAB thin-walled beam analysis tool for wingbox stress, shear flow, and twist prediction.
- Conducted composite-versus-aluminum trade studies demonstrating substantial structural mass savings.

EXPERIENCE

Design-Build-Fly (DBF) — Aerodynamics & Structures — UC San Diego | Sep 2025 – June 2026

- Led airfoil selection, aerodynamic trade studies, and structural sizing for a competition RC aircraft.

ITS Service Desk — Business and Technical Support Analyst — UC San Diego | Sep 2023 – May 2026

- Directed a 60+ member technician team providing Tier-1 IT support and technical troubleshooting.
- Founded a zero-tier support social media channel to streamline issue resolution and reduce ticket volume.

LEADERSHIP & ACTIVITIES

- **International Student Society** — Event Coordinator & Recruitment Chair; mentored 50+ international students.
- **Phi Gamma Delta** — Scholarship & Graduate Relations Chair; organized professional development and alumni networking events.
- **Duke of Edinburgh Gold Award** — Leadership, community service, and expedition achievement.